TRAINERS AND WHEELS · 12 MIN READ

Wheels, tyres, rotors and freehubs: choosing the right wheelset

A wheelset is the biggest upgrade a customer can make to a stock bike. It is also the easiest part to sell wrongly, because a wheel has to match the frame, the brakes, the drivetrain and the tyres. This lesson gives you the checks in order.

1. What a wheel is made of

- **Rim**: aluminium (alloy) or carbon. Alloy is cheaper and tougher. Carbon is lighter for the same depth and can be made deep and aero.
- **Hub**: the centre. It holds the axle, the bearings, the brake rotor mount and, on the rear, the freehub body.
- **Spokes**: connect the two.
- **Freehub body**: the splined part of the rear hub that the cassette slides onto.

2. Wheel sizes

- **700c**: road, gravel and hybrid. A 29 inch mountain bike wheel has the same rim diameter.
- **650b**: the same rim diameter as a 27.5 inch mountain bike wheel. Used on some gravel bikes and smaller frames.

3. Brake type: the first check

A wheel is built for rim brakes or for disc brakes, never both. A rim brake wheel has a braking surface on the rim and no rotor mount. A disc wheel has a rotor mount on the hub and no braking surface. We list both: for example the Token Konax Pro 52 comes in a rim brake version and a disc version. Ask before anything else.

4. Rotors

TOPIC	WHAT TO KNOW
Mount	**Centerlock**: the rotor slides onto splines and one lockring holds it. **6-bolt**: six small bolts hold it. A 6-bolt rotor can go on a Centerlock hub with an adapter. A Centerlock rotor cannot go on a 6-bolt hub.
Size	Road: 140 or 160 mm. Mountain bike: 160, 180 or 203 mm. Bigger rotors give more braking power and cope better with heat on long descents.
Limits	The frame and fork set the smallest and largest rotor that fits. Check before changing size.

5. Axles

- **Quick release**: a thin skewer with a lever. Found on rim brake bikes and older or entry disc bikes.
- **Thru-axle**: a thick axle that threads into the frame. Road and gravel disc: 12 × 100 mm front, 12 × 142 mm rear. Mountain bike Boost: 15 × 110 mm front, 12 × 148 mm rear.

The listing tells you. "Mavic Cosmic SL 45 Disc, Centerlock 12×100/12×142" means disc brake, Centerlock rotors, road thru-axles. A road wheel will not fit a Boost mountain bike frame.

6. Freehub bodies: Shimano and SRAM

The cassette decides the freehub body. Get this wrong and the customer's cassette will not go onto the new wheel.

FREEHUB BODY	TAKES THESE CASSETTES
Shimano HG	Shimano 8 to 11 speed, Shimano 12-speed road (105, Ultegra, Dura-Ace), and SRAM cassettes whose smallest cog is 11 teeth.
SRAM XDR	SRAM 12-speed AXS road cassettes, which start at a 10-tooth cog.
SRAM XD	SRAM Eagle mountain bike cassettes that start at 10 teeth. XD is slightly shorter than XDR. An XD cassette fits an XDR body with a spacer.
Shimano Micro Spline	Shimano 12-speed mountain bike cassettes.

The quick test: a cassette with a **10-tooth** smallest cog never fits a Shimano HG body. On most hubs the freehub body can be swapped, and we stock XDR bodies for several brands. Check that one is available for that wheel before you promise it.

7. Tyre systems

- **Clincher**: tyre plus inner tube. Simple, and easy to fix at the roadside.
- **Tubeless**: no tube. An airtight tyre, rim tape, a tubeless valve and liquid sealant. It runs at lower pressure for more grip and comfort, and the sealant closes small punctures. It needs a tubeless-ready rim and a tubeless tyre. "Tubeless Ready" or "UST" in a wheel's name tells you the rim is.
- **Tubular**: the tyre is glued to the rim. Used on the track and by some racers. Not for everyday riders.

Hookless rims: some carbon rims have straight inner walls with no hook. They must be used with tubeless tyres approved for hookless rims, and at no more than 5 bar (72.5 psi). Tell the customer both points.

Width: modern rims are wider inside, about 21 to 25 mm, which suits 28 to 32 mm tyres. Check the frame's tyre clearance before selling a wider tyre.

8. Rim depth

DEPTH	GOOD FOR	TRADE-OFF
Shallow, under 35 mm	Climbing, light riders, windy days	Least aero
Mid, 40 to 50 mm	All-round riding. The best single wheelset for most riders.	A little heavier than shallow
Deep, 55 mm and more	Flat roads, time trials, triathlon	Heavier, and crosswinds push the front wheel

A deeper rim cuts through air better, so it is faster on flat roads at speed. It also weighs more and catches crosswinds, and a light rider feels that most. That is why some sets are mixed: the Magene EXAR DB508 is 50 mm front and 58 mm rear. The shallower front wheel keeps the steering calm, and the deeper rear adds speed where wind matters less.

9. Our range at a glance

USE	WHEELSETS	LISTED FROM
Alloy, training and replacement	DT Swiss P1850 Spline, Shimano RS171, Token Resolute C22A (rim brake), Gist CL Alloy 42	₹17,000
Carbon, mid depth	Elite Wheels Marvel 50D, Magene EXAR Pro DB45, Token Resolute C45D, Mavic Cosmic SL 45	₹70,999
Carbon, deep	Magene EXAR Ultra DB508, Elite Wheels Drive Helix 57D, Vision SC 55 and SC 60	₹1,05,900
Gravel	QUAI G50	₹48,500
Mountain bike	Token Prime 29 inch Boost	₹39,390
Track	Token Resolute Track, Mavic Ellipse Pro Carbon	₹26,990

Prices are our website listings in November 2026. They change, so check the site before you quote.

Do not confuse two brands: **Elite Wheels** makes carbon wheelsets. **Elite** is the Italian company that makes indoor trainers. They are not the same company.

10. Choosing the wheel: the order to ask

- **Brake**: rim or disc? If disc, Centerlock or 6-bolt rotors?
- **Axle**: quick release or thru-axle, and which size?
- **Drivetrain**: which cassette, so which freehub body?
- **Tyres**: what width fits the frame, and does the customer want tubeless?
- **Riding**: hills or flat, racing or training, how windy, how heavy is the rider?
- **Budget**.

The first four decide what *fits*. The last two decide what is *best*. Never skip to depth and price before the fit questions are answered.

11. Matching wheel to rider

- Rides ghats every weekend, 60 kg: shallow to mid depth, around 35 to 45 mm.
- One wheelset for everything: 40 to 50 mm.
- Flat-road racer or triathlete: 55 mm and deeper, or a mixed-depth set.
- Daily training and bad roads: a tough alloy set, and keep the carbon set for events.

With every wheelset, check whether the customer also needs rotors, a cassette, tubeless valves and sealant, and rim brake pads for carbon rims if the wheels are rim brake.

Remember these

- Fit first: brake type, axle, freehub, tyre. Then depth and budget.
- Rim brake and disc wheels are never interchangeable. Centerlock hubs take 6-bolt rotors with an adapter, not the other way round.
- A 10-tooth smallest cog never fits Shimano HG. SRAM AXS road needs XDR.
- Mid depth, 40 to 50 mm, is the best all-round choice. Deep is for flat speed and costs weight and crosswind stability.
- Hookless rims: approved tubeless tyres only, 5 bar maximum.